

Black holes have played an important part in astrophysics for the last half-century. A black hole is a region of space where the force of gravity is especially strong and gravitational potential energy U is very large. Analogous to the electrostatic potential energy associated with electrical charges, U is defined as the work required to move an object of mass m from an infinite distance to a point at radial distance r within the gravitational field of a second mass M as given by

$$U = - \frac{G M m}{r}$$

Equation 1

where G is the gravitational constant. Near a black hole, the gravitational potential energy can become so negative that not even light has enough energy to escape the black hole's gravity.

Escape velocity can be determined by launching an object of mass m with velocity v to an infinite radial distance from the surface of a central mass M (eg, a star or a planet) with radius R . For a black hole to exist, the escape velocity must be equal to the speed of light.

Scientists have thoroughly studied the physics of black holes. Stars near the center of our galaxy have been observed to orbit an extremely massive, invisible object (millions of times more massive than our Sun) that is almost certainly a black hole. The orbital angular velocity ω of a star is defined as

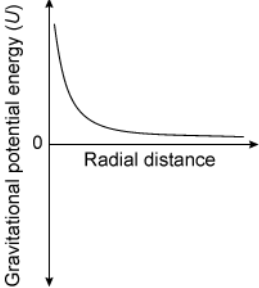
$$\omega = 360^\circ \cdot f$$

Equation 2

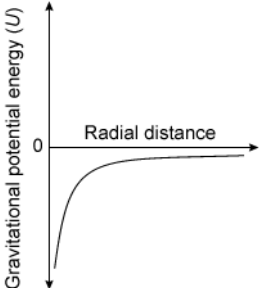
where f is the frequency of rotation in hertz (Hz). Likewise, black holes are not merely static masses but are often observed to rotate about their own axis. The interaction between the rotation of a black hole and matter falling into the hole is suspected to form powerful beams (jets) of ionized matter that project away from the black hole along its axis of rotation. Nevertheless, many aspects of how these jets are formed remains a mystery. Black holes will remain a fruitful area for research far into the future.

Which of the following graphs best illustrates how the gravitational potential energy for a black hole varies with radial distance r ?

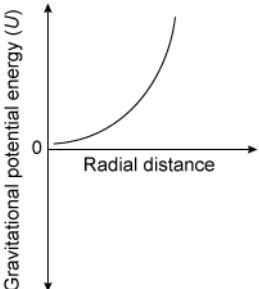
- ☐ A.



[10%]
- ☒ B.

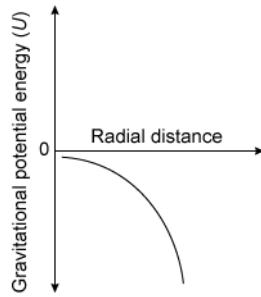


[69%]
- ☐ C.



[3%]

☐ D.



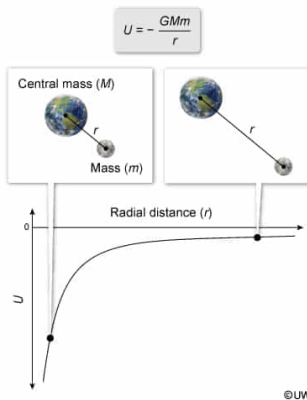
[16%]

Correct

68% answered correctly

Explanation:

Gravitational potential energy (U)



The total energy within a mechanical system is the sum of the system's kinetic and potential energy. Kinetic energy is derived from the motion of objects within a system whereas **potential energy** depends on the position of objects within the system. Because potential energy is related to position and not path, potential energy is associated with **conservative forces** (eg, gravitational, electrostatic, elastic forces).

When establishing a reference point to be used for calculating potential energy, an infinite radial distance is chosen as a convenient location at which the potential energy is zero. Because potential energy measures the potential of conservative forces to do work, the change in potential energy (ΔU) relative to this point is equal and opposite to the work (W) done by conservative forces within the system:

$$\Delta U = -W$$

Equation 1 in the passage shows that gravitational potential energy (U) is **inversely proportional** to the radial distance (r) separating the object mass (m) from the central mass (M) (eg, a black hole). The chosen reference point is an infinite radius where U is zero and the work done by gravity in moving an object from infinity closer to the central mass is **positive**. Therefore, ΔU is negative and the potential energy at a finite radius (U_r) is always negative:

$$\Delta U = U_r - U_{\text{infinity}} = U_r - 0 < 0$$

$$U_r < 0$$

Because U_r is always negative, the curve of inverse proportionality is located within the bottom-right quadrant of an xy -coordinate system. Therefore, the graph that best fits the trend of Equation 1 is an inversely proportional plot located below the x -axis (**Choice B**).

(Choice A) This graph depicts an inverse relationship between U and r . However, Equation 1 includes a negative sign on the right side of the equation, meaning that U is always negative rather than positive.

(Choices C and D) These graphs depict large (negative or positive) potential energy at large radial distances and zero potential energy at $r = 0$. Therefore, both are inconsistent with an inversely proportional relationship.

Educational objective:

Gravitational potential energy measures the potential of gravity to do work on a mass. Conventionally, gravitational potential energy is expressed using negative numbers and is inversely proportional to the radial distance between masses.

Time spent: 0 sec
Subject:
Physics

QID: 402698
System:
Mechanics and Energy

Black holes have played an important part in astrophysics for the last half-century. A black hole is a region of space where the force of gravity is especially strong and gravitational potential energy U is very large. Analogous to the electrostatic potential energy associated with electrical charges, U is defined as the work required to move an object of mass m from an infinite distance to a point at radial distance r within the gravitational field of a second mass M as given by

$$U = - \frac{G M m}{r}$$

Equation 1

where G is the gravitational constant. Near a black hole, the gravitational potential energy can become so negative that not even light has enough energy to escape the black hole's gravity.

Escape velocity can be determined by launching an object of mass m with velocity v to an infinite radial distance from the surface of a central mass M (eg, a star or a planet) with radius R . For a black hole to exist, the escape velocity must be equal to the speed of light.

Scientists have thoroughly studied the physics of black holes. Stars near the center of our galaxy have been observed to orbit an extremely massive, invisible object (millions of times more massive than our Sun) that is almost certainly a black hole. The orbital angular velocity ω of a star is defined as

$$\omega = 360^\circ \cdot f$$

Equation 2

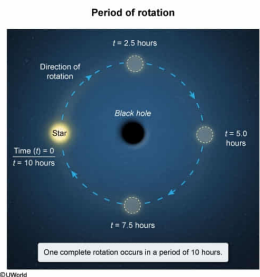
where f is the frequency of rotation in hertz (Hz). Likewise, black holes are not merely static masses but are often observed to rotate about their own axis. The interaction between the rotation of a black hole and matter falling into the hole is suspected to form powerful beams (jets) of ionized matter that project away from the black hole along its axis of rotation. Nevertheless, many aspects of how these jets are formed remains a mystery. Black holes will remain a fruitful area for research far into the future.

A star orbiting a supermassive black hole makes two approximately circular revolutions around the black hole every 20 hours. What is the approximate angular velocity of the star?

- ☐ A. 5.0×10^{-3} degrees/second [17%]
- ☒ B. 1.0×10^{-2} degrees/second [61%]
- ☐ C. 3.0×10^{-1} degrees/second [14%]
- ☐ D. 6.0×10^{-1} degrees/second [5%]

Correct
62% answered correctly

Explanation:



Periodic motion occurs when an object **moves in a cyclical fashion**, returning to the same location after a certain period of time. Classic examples of periodic motion are the oscillatory behavior of a clock pendulum, or a mass located on the end of a perturbed spring.

Periodic motion is modeled in terms of **wave motion**, where the time required to complete **one cycle of motion** is defined as the **period (T)**. The frequency (f) of oscillations per interval of time, which is often expressed in units of cycles per second (hertz), is the inverse of the period:

$$f = \frac{1}{T}$$

The orbital motion of a mass around a central body is also periodic. In this question, a star orbiting around a supermassive black hole completes two revolutions in 20 hours. Consequently, T within this system is equal to 10 hours. Using this period to calculate the frequency and converting the units from hours to seconds yields:

$$f = \frac{1}{10 \text{ hours}} = \frac{1 \text{ cycle}}{10 \text{ hours}}$$

Using the period (expressed in seconds) determined from one complete orbital cycle, Equation 2 may be used to determine the angular velocity of the orbit:

$$\omega = 360^\circ \cdot f = 360^\circ \cdot \left(\frac{1}{36,000 \text{ s}} \right) = 1.0 \times 10^{-2} \frac{\text{degrees}}{\text{second}}$$

Therefore, the angular velocity of the star orbiting the supermassive black hole is 1.0×10^{-2} degrees/second (**Choice B**).

Educational objective:

Periodic motion describes an object moving in a cyclical fashion in which the same movements recur in regular intervals. The frequency of cyclical motion is the inverse of period, which is the time required to complete one cycle.

Black holes have played an important part in astrophysics for the last half-century. A black hole is a region of space where the force of gravity is especially strong and gravitational potential energy U is very large. Analogous to the electrostatic potential energy associated with electrical charges, U is defined as the work required to move an object of mass m from an infinite distance to a point at radial distance r within the gravitational field of a second mass M as given by

$$U = - \frac{G M m}{r}$$

Equation 1

where G is the gravitational constant. Near a black hole, the gravitational potential energy can become so negative that not even light has enough energy to escape the black hole's gravity.

Escape velocity can be determined by launching an object of mass m with velocity v to an infinite radial distance from the surface of a central mass M (eg, a star or a planet) with radius R . For a black hole to exist, the escape velocity must be equal to the speed of light.

Scientists have thoroughly studied the physics of black holes. Stars near the center of our galaxy have been observed to orbit an extremely massive, invisible object (millions of times more massive than our Sun) that is almost certainly a black hole. The orbital angular velocity ω of a star is defined as

$$\omega = 360^\circ \cdot f$$

Equation 2

where f is the frequency of rotation in hertz (Hz). Likewise, black holes are not merely static masses but are often observed to rotate about their own axis. The interaction between the rotation of a black hole and matter falling into the hole is suspected to form powerful beams (jets) of ionized matter that project away from the black hole along its axis of rotation. Nevertheless, many aspects of how these jets are formed remains a mystery. Black holes will remain a fruitful area for research far into the future.

An object is launched with a one-time burst of propulsion away from the surface of the Moon. After the burst, which of the following best describes the changes that occur as the object moves away from the Moon's surface?

- ☐ A. The object mass is dissipated as heat.
- ☐ B. Potential energy is converted into kinetic energy. [38%]
- ☒ C. Kinetic energy is converted into potential energy. [55%]
- ☐ D. Total mechanical energy is not conserved. [4%]

Correct
54% answered correctly

Explanation:

Conservation of energy (object launched in gravity)

Change in kinetic energy (ΔKE) + change in potential energy (ΔPE) = 0

© UWisc

The work-energy principle states that the total work W done by a force is equal to the change in kinetic energy (ΔKE) of the object acted upon by the force:

$$W = \Delta K E$$

If the forces acting on the object are conservative forces (eg, gravity), then the work done by the force also contributes to a change in the potential energy of the system (ΔPE):

$$W = - \Delta P E$$

Combining these two equations yields a statement expressing the conservation of energy in mechanical systems:

$$\Delta K E + \Delta P E = 0$$

Furthermore, the above equation may be restated in terms of the initial (i) and final values (f) of each variable:

$$(K E_f - K E_i) + (P E_i - P E_f) = 0$$

Consequently, within purely mechanical systems any change in the kinetic energy is accompanied by an equal and opposite change in potential energy such that total mechanical energy remains the same in both conditions.

After the object (eg, a rocket) is launched from the Moon, the only force present in the system is gravity. As the rocket moves away, the gravitational force is oriented toward the Moon. Therefore, work done by gravity progressively decreases the velocity and the kinetic energy of the object:

$$K E_i > K E_f$$

Simultaneously, the potential energy of the object increases (becomes less negative) as the radial distance from the mass increases:

$$P E_i < P E_f$$

The decrease in the kinetic energy of the object is accompanied by an increase in the potential energy of the object such that conservation of energy is maintained.

(Choice A) The mass of the object is independent of gravity, and no forces act on the object that would cause depletion of mass.

(Choice B) After launch, the PE of the object increases while the KE of the object decreases. Therefore, PE could not have been transformed into KE .

(Choice D) The total mechanical energy of the system is conserved because gravity is the only force acting on the object after it is launched (no drag force is present because the Moon has no atmosphere).

Educational objective:

Conservation of energy states that any change in kinetic energy is accompanied by an equal and opposite change in potential energy such that the total mechanical energy remains constant. The kinetic energy of an object launched upward in gravity is converted into gravitational potential energy.

Black holes have played an important part in astrophysics for the last half-century. A black hole is a region of space where the force of gravity is especially strong and gravitational potential energy U is very large. Analogous to the electrostatic potential energy associated with electrical charges, U is defined as the work required to move an object of mass m from an infinite distance to a point at radial distance r within the gravitational field of a second mass M as given by

$$U = - \frac{G M m}{r}$$

Equation 1

where G is the gravitational constant. Near a black hole, the gravitational potential energy can become so negative that not even light has enough energy to escape the black hole's gravity.

Escape velocity can be determined by launching an object of mass m with velocity v to an infinite radial distance from the surface of a central mass M (eg. a star or a planet) with radius R . For a black hole to exist, the escape velocity must be equal to the speed of light.

Scientists have thoroughly studied the physics of black holes. Stars near the center of our galaxy have been observed to orbit an extremely massive, invisible object (millions of times more massive than our Sun) that is almost certainly a black hole. The orbital angular velocity ω of a star is defined as

$$\omega = 360^\circ \cdot f$$

Equation 2

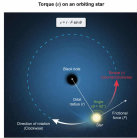
where f is the frequency of rotation in hertz (Hz). Likewise, black holes are not merely static masses but are often observed to rotate about their own axis. The interaction between the rotation of a black hole and matter falling into the hole is suspected to form powerful beams (jets) of ionized matter that project away from the black hole along its axis of rotation. Nevertheless, many aspects of how these jets are formed remains a mystery. Black holes will remain a fruitful area for research far into the future.

A star orbiting a black hole in the clockwise direction begins to slowly spiral inward due to a counterclockwise drag force. When the star's radial distance from the black hole is 1.0×10^6 km, a drag force of 250 N acts perpendicular to the orbit radius. What is the torque on the star?

- ☐ A. 0 N·m [8%]
- ☐ B. 2.5×10^6 N·m in the counterclockwise direction [17%]
- ☐ C. 2.5×10^{11} N·m in the clockwise direction [21%]
- ☒ D. 2.5×10^{11} N·m in the counterclockwise direction [52%]

Correct
53% answered correctly

Explanation:



A **torque (τ)** is a rotational force caused by a translational force (F) applied at some radial distance (r) from a center of rotation. Torques can cause rotation about a central axis or oppose any rotation already present. The magnitude of a torque is defined as:

$$\tau = r \cdot F \cdot \sin \theta$$

where θ is the angle between the translational force and the radial vectors. When r and F are perpendicular ($\theta = 90^\circ$), the torque is maximal:

$$\tau = r \cdot F \cdot \sin (90^\circ)$$

Conversely, when r and F are parallel ($\theta = 0^\circ$), the magnitude of torque is zero (the force has no rotational influence).

In this question, a star orbits a black hole in a **clockwise direction** at a radial distance of 1.0×10^6 km. The translational force applied on the star acts in a **counterclockwise direction** (opposite to the clockwise orbit of the star), perpendicular to the radial vector. Consequently, θ is 90° and the magnitude of the torque is:

$$\tau = (1.0 \times 10^6 \text{ km}) \cdot (250 \text{ N})$$

Converting km to m ($1 \text{ km} = 1000 \text{ m}$) yields:

$$\tau = (1.0 \times 10^9 \text{ m}) \cdot (250 \text{ N}) = 2.5 \times 10^{11} \text{ N} \cdot \text{m}$$

Therefore, the star is subjected to a counterclockwise torque (by the **right-hand rule** counterclockwise corresponds to a torque vector directed out of the page) with a magnitude of 2.5×10^{11} N·m (**Choice D**).

Educational objective:

Torques are forces that influence the rotational motion of a system in a clockwise or counterclockwise direction. The magnitude of a torque is the product of the applied translational force, the radial distance from the center of rotation, and the sine of the angle between the translational force and radial vectors.

Time spent: 0 sec
Subject:
Physics

QID: 462701
System:
Mechanics and Energy

Black holes have played an important part in astrophysics for the last half-century. A black hole is a region of space where the force of gravity is especially strong and gravitational potential energy U is very large. Analogous to the electrostatic potential energy associated with electrical charges, U is defined as the work required to move an object of mass m from an infinite distance to a point at radial distance r within the gravitational field of a second mass M as given by

$$U = - \frac{G M m}{r}$$

Equation 1

where G is the gravitational constant. Near a black hole, the gravitational potential energy can become so negative that not even light has enough energy to escape the black hole's gravity.

Escape velocity can be determined by launching an object of mass m with velocity v to an infinite radial distance from the surface of a central mass M (eg, a star or a planet) with radius R . For a black hole to exist, the escape velocity must be equal to the speed of light.

Scientists have thoroughly studied the physics of black holes. Stars near the center of our galaxy have been observed to orbit an extremely massive, invisible object (millions of times more massive than our Sun) that is almost certainly a black hole. The orbital angular velocity ω of a star is defined as

$$\omega = 360^\circ \cdot f$$

Equation 2

where f is the frequency of rotation in hertz (Hz). Likewise, black holes are not merely static masses but are often observed to rotate about their own axis. The interaction between the rotation of a black hole and matter falling into the hole is suspected to form powerful beams (jets) of ionized matter that project away from the black hole along its axis of rotation. Nevertheless, many aspects of how these jets are formed remains a mystery. Black holes will remain a fruitful area for research far into the future.

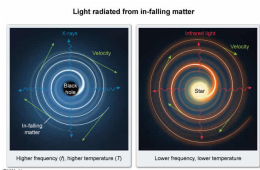
Matter falling in the gravitational field of a central mass can form a hot, charged fluid. The fluid matter orbiting some stars radiates infrared light, but the matter orbiting black holes emits x-rays. Which of the following explains this discrepancy? The matter orbiting black holes must:

- ✔ ☒ A. be hotter than the matter orbiting stars. [56%]
- ☐ B. be colder than the matter orbiting stars. [13%]
- ☐ C. have lower kinetic energy than the matter orbiting stars. [24%]
- ☐ D. exhibit higher viscosity than the matter orbiting stars. [4%]

Correct

57% answered correctly

Explanation:



Electromagnetic (EM) waves are prevalent throughout nature in a variety of different frequencies and wavelengths. EM waves are arranged by frequency (or wavelength) into the EM spectrum, which ranges from very low frequency (long wavelength) radio waves to very high frequency (small wavelength) gamma rays.

When EM waves are emitted or absorbed, they behave as photons. Photon energy (E) is related to Planck's constant (h) and waveform frequency (f):

$$E = h \cdot f$$

Therefore, relatively high-frequency EM radiation (eg, gamma rays) are more energetic whereas relatively low-frequency EM radiation (eg, radio waves) carries much less energy.

Radiation energy is directly related to the temperature of the object that emits the radiation. Hotter objects radiate waves in the higher frequency region of the spectrum; conversely, colder objects radiate waves in the lower frequency region. For example, the human body radiates infrared light (ie, with frequencies just below those of visible red light).

The matter orbiting black holes emits x-rays, which have a higher frequency and a higher energy than the infrared radiation emitted by matter orbiting stars:

$$f_{\text{infrared}} < f_{\text{x-rays}} \\ E_{\text{infrared}} < E_{\text{x-rays}}$$

Because the energy of x-ray radiation is greater than that of infrared radiation, the temperature of the matter orbiting black holes must be greater than the temperature of the matter orbiting the stars.

(Choice B) Infrared radiation is less energetic than x-rays. Therefore, the matter orbiting black holes cannot be colder than the matter orbiting the stars because warmer bodies emit radiation with higher relative energy.

(Choice C) If matter orbiting the stars had a higher kinetic energy than the matter orbiting black holes, the matter around the stars would emit a form of radiation with higher frequencies than x-rays.

(Choice D) The viscosity of the in-falling matter is a property of the fluid itself and is not related to either the central mass of the system (ie, black hole, star) or the light emitted by the fluid.

Educational objective:

The electromagnetic (EM) spectrum classifies EM waves by frequency or wavelength. Objects with high energy have a higher temperature and emit EM waves with relatively higher frequencies and energy.

Time spent:0 sec

Subject: Physics

QID:402702

System: Mechanics and Energy

Black holes have played an important part in astrophysics for the last half-century. A black hole is a region of space where the force of gravity is especially strong and gravitational potential energy U is very large. Analogous to the electrostatic potential energy associated with electrical charges, U is defined as the work required to move an object of mass m from an infinite distance to a point at radial distance r within the gravitational field of a second mass M as given by

$$U = - \frac{G M m}{r}$$

Equation 1

where G is the gravitational constant. Near a black hole, the gravitational potential energy can become so negative that not even light has enough energy to escape the black hole's gravity.

Escape velocity can be determined by launching an object of mass m with velocity v to an infinite radial distance from the surface of a central mass M (eg, a star or a planet) with radius R . For a black hole to exist, the escape velocity must be equal to the speed of light.

Scientists have thoroughly studied the physics of black holes. Stars near the center of our galaxy have been observed to orbit an extremely massive, invisible object (millions of times more massive than our Sun) that is almost certainly a black hole. The orbital angular velocity ω of a star is defined as

$$\omega = 360^\circ \cdot f$$

Equation 2

where f is the frequency of rotation in hertz (Hz). Likewise, black holes are not merely static masses but are often observed to rotate about their own axis. The interaction between the rotation of a black hole and matter falling into the hole is suspected to form powerful beams (jets) of ionized matter that project away from the black hole along its axis of rotation. Nevertheless, many aspects of how these jets are formed remains a mystery. Black holes will remain a fruitful area for research far into the future.

A black hole emits a jet with a power of 10^{44} W. If the black hole has a rotational period of 0.1 ms, what is the energy emitted by the jet during one rotation of the black hole?

-
- A. 10^{40} J [62%]

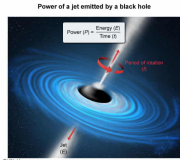
B. 10^{43} J [13%]

C. 10^{45} J [8%]

D. 10^{48} J [14%]

Correct
62% answered correctly

Explanation:



Power (P) measures the rate at which **energy (E)** is expended by a physical process over a **period of time (t)**:

$$P = \frac{E}{t}$$

In mechanics, **power** is the amount work done by a force acting on a system per unit time. Power can also reflect a **rate of energy transfer** or conversion between systems. For example, **electric power** in a circuit is the rate at which energy is dissipated as heat and light by an electric current passing through a resistor.

In the case of a black hole, the immense rotational and magnetic forces acting on nearby charged matter cause a mechanical acceleration of the matter, which forms high-energy jets that travel away from the black hole along its axis of rotation. The power associated with these jets is:

$$P = 10^{44} \text{ W} = 10^{44} \frac{\text{J}}{\text{s}}$$

The energy emitted by the jet during one rotation of the black hole is related to the power of the jets and the rotational period of the black hole:

$$E = 10^{44} \frac{\text{J}}{\text{s}} \cdot \left(0.1 \text{ ms} \cdot \frac{1 \text{ s}}{1000 \text{ ms}} \right)$$
$$E = 10^{44} \text{ J} \cdot \left(1 \times 10^{-4} \right)$$
$$E = 10^{40} \text{ J}$$

Therefore, the amount of energy released by the jet during one rotation of the black hole is 10^{40} J (**Choice A**).

Educational objective:

Power describes the rate at which energy is expended within a system per unit time. Power also quantifies the transfer of energy or the total work done on a system over a period of time.

Time spent: 0 sec
Subject:
Physics

QID:402703
System:
Mechanics and Energy